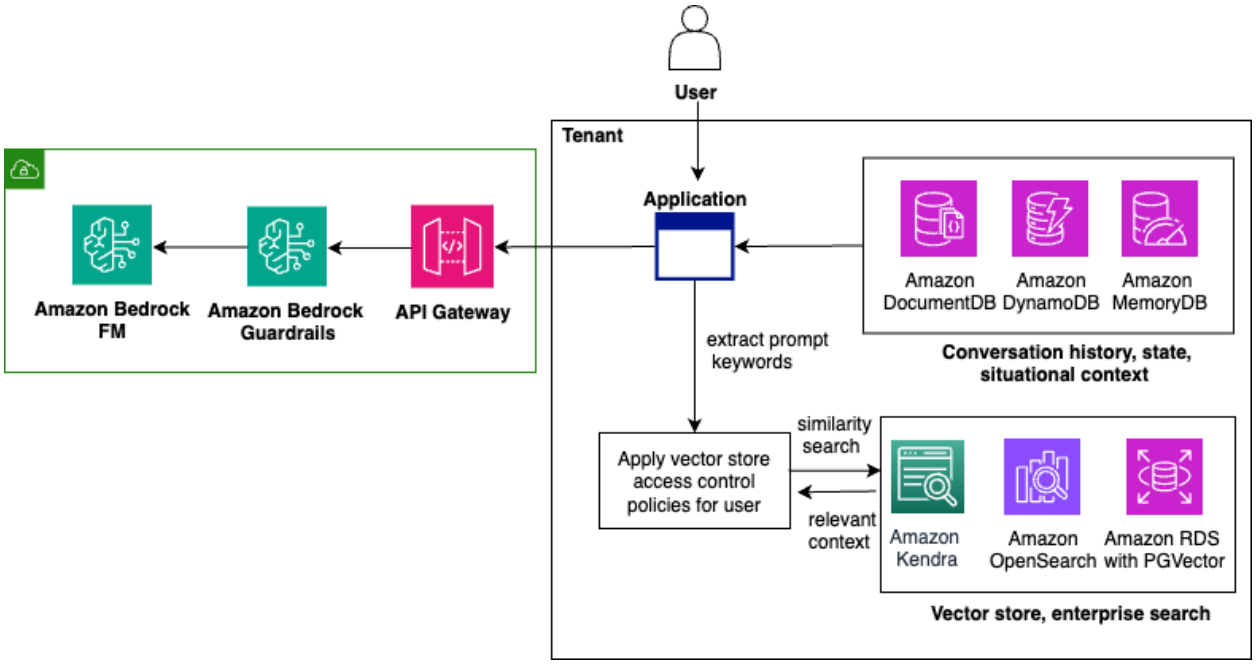
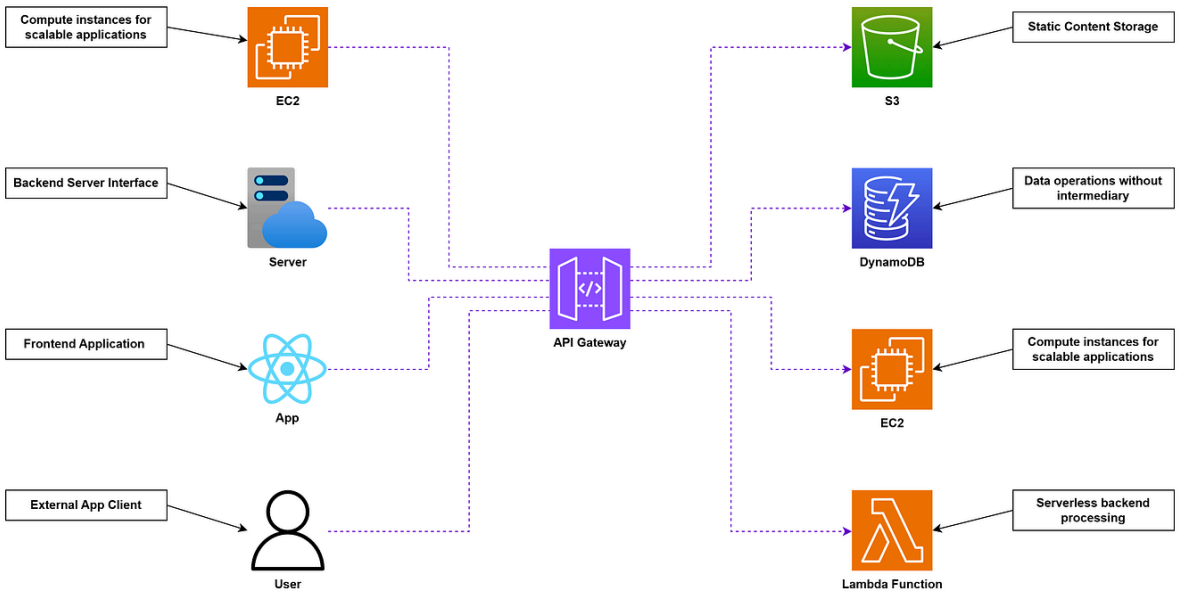
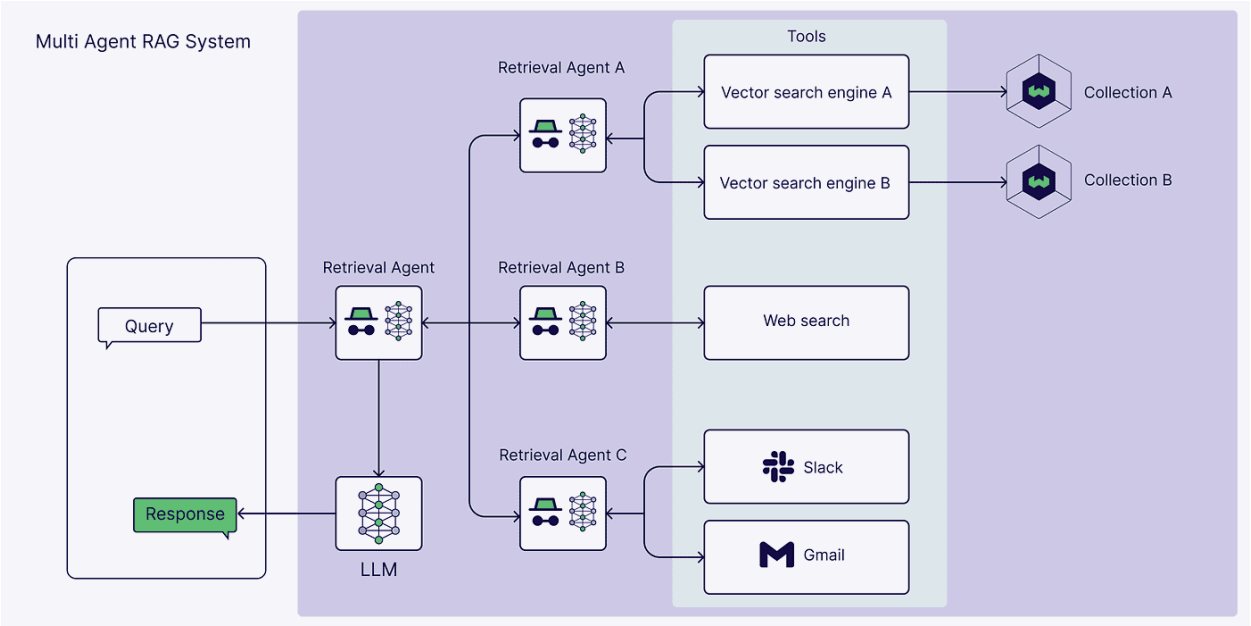
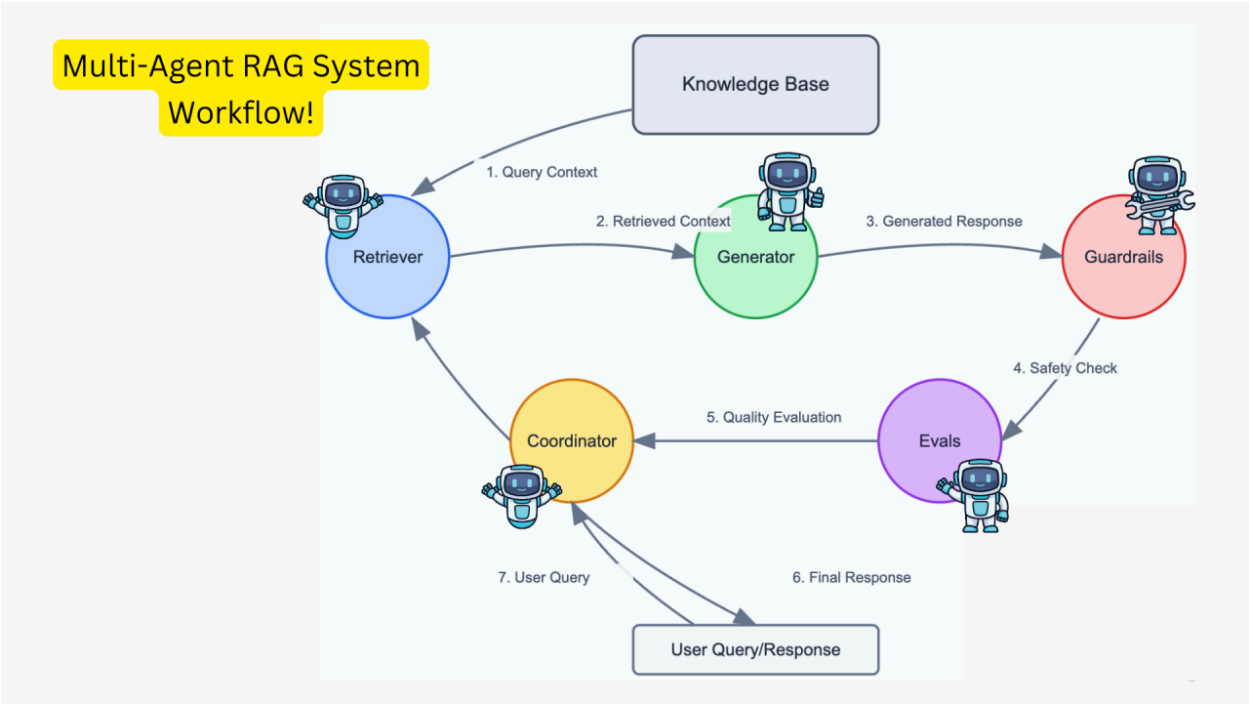
Final Enterprise Architecture

- Fully private VPC (no internet)
- Multi-agent AI system
- RAG with vector DB
- Bedrock LLM via VPC endpoint
- CI/CD with GitHub OIDC
- Blue/Green deployment with auto rollback
- Multi-environment (dev/staging/prod)




Compute instances for scalable applications
EC2
Backend Server Interface
Server
Frontend Application
App
External App Client
User
API Gateway
Static Content Storage
S3
Data operations without intermediary
DynamoDB
Compute instances for scalable applications
EC2
Serverless backend processing
Lambda Function


Multi Agent RAG System
Query
Response
Retrieval Agent
LLM
Retrieval Agent A
Retrieval Agent B
Retrieval Agent C
Tools
Vector search engine A
Vector search engine B
Web search
Slack
Gmail
Collection A
Collection B



Architecture Explanation

🌐 Entry Layer

- **API Gateway**
  - Public entry point
  - Routes `/agent` requests
  - Can be extended with JWT auth

⚙️ Compute Layer (Private)

- **Lambda (inside VPC)**
  - Multi-agent execution:

        - Planner
        - Retriever (RAG)
        - Executor
        - Critic
    - No internet access

🧠 AI Layer

- **Amazon Bedrock**
    - Claude (reasoning)
    - Titan (embeddings)
    - Access via **VPC Endpoint**

🔎 RAG Layer

- **OpenSearch (Vector DB)**
    - Private access only
    - Semantic search
    - Stores embeddings

🗄️ Data Layer

- **DynamoDB**
    - session memory
- **S3 (optional)**
    - document storage

🔐 Security Layer

- **VPC (Private Subnets)**
- **Security Groups**

- **Secrets Manager**
- **IAM (least privilege)**
- **OIDC (GitHub → AWS)**

🔁 Deployment Layer

- **CodeDeploy**
  - Blue/Green
  - Canary rollout (10%)
  - Auto rollback (CloudWatch alarm)

📊 Observability

- **CloudWatch**
  - logs + metrics
- **X-Ray**
  - tracing
- **Alarms**
  - trigger rollback

🔁 CI/CD Pipeline

- **GitHub Actions**
  - OIDC authentication
  - Terraform deploy
  - CodeDeploy trigger

🌍 Multi-Environment

- dev / staging / prod
- isolated via:

    - `tfvars`
    - naming (`-dev`, `-prod`)
    - CI/CD environments

🔄 End-to-End Flow

User → API Gateway

↓

Lambda (VPC)

↓

├ RAG → OpenSearch (vector search)

├ LLM → Bedrock (via VPC endpoint)

├ Memory → DynamoDB

└ Config → Secrets Manager

↓

Response

CI/CD:

GitHub → OIDC → Terraform → CodeDeploy → Lambda Alias

Deployment Flow:

Git Push

↓

GitHub Actions

↓

OIDC → AWS

↓

Terraform Apply

↓

New Lambda Version

↓

CodeDeploy (10% traffic)

↓

Monitor

↓

Auto rollback OR full rollout

🔐 What is OIDC in our Setup

Instead of storing secrets:

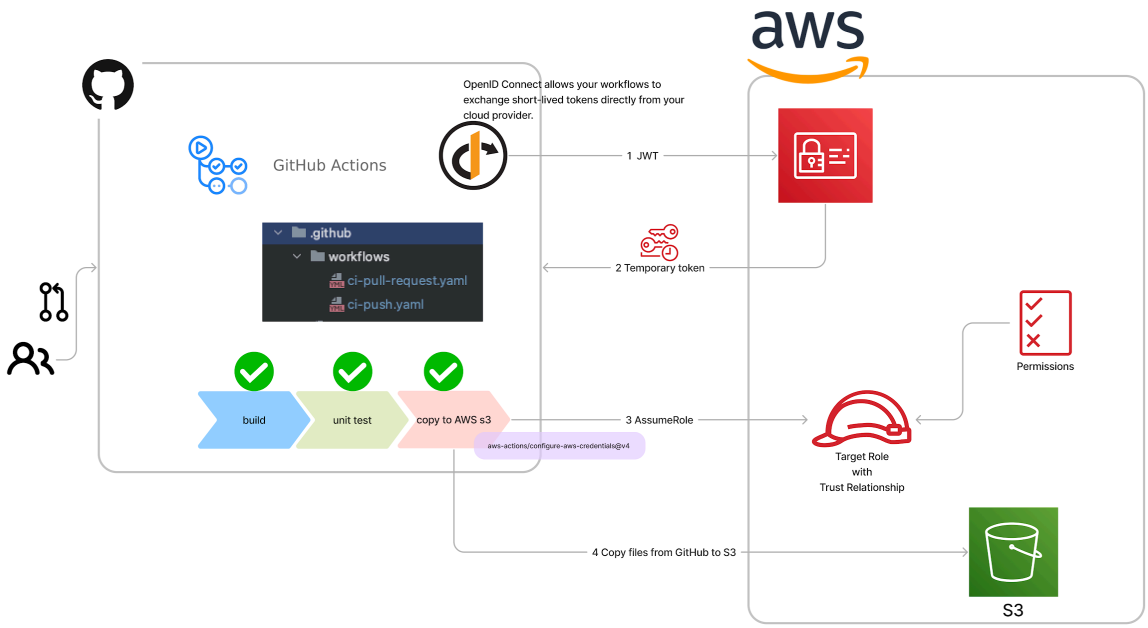
GitHub → AWS using ACCESS KEYS ❌

You do:

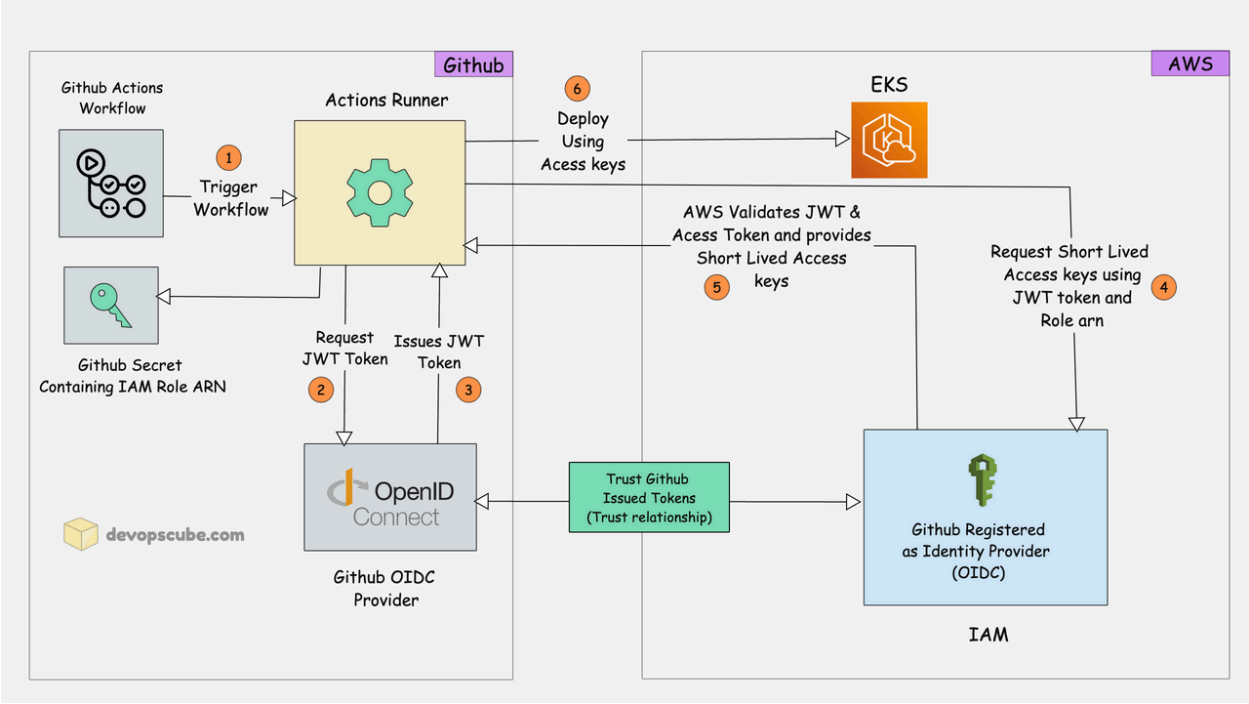
GitHub → OIDC token → Assume IAM Role → AWS ✅

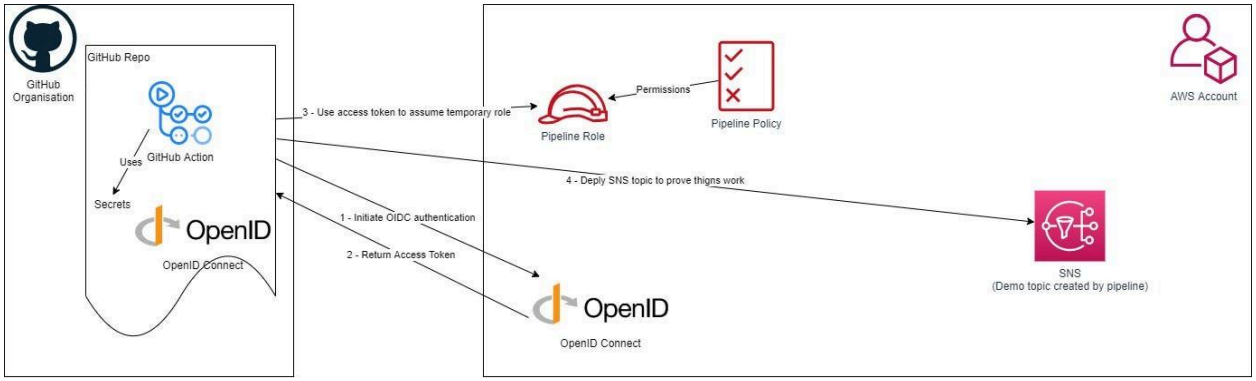
👉 No secrets stored
👉 Temporary credentials
👉 Much safer

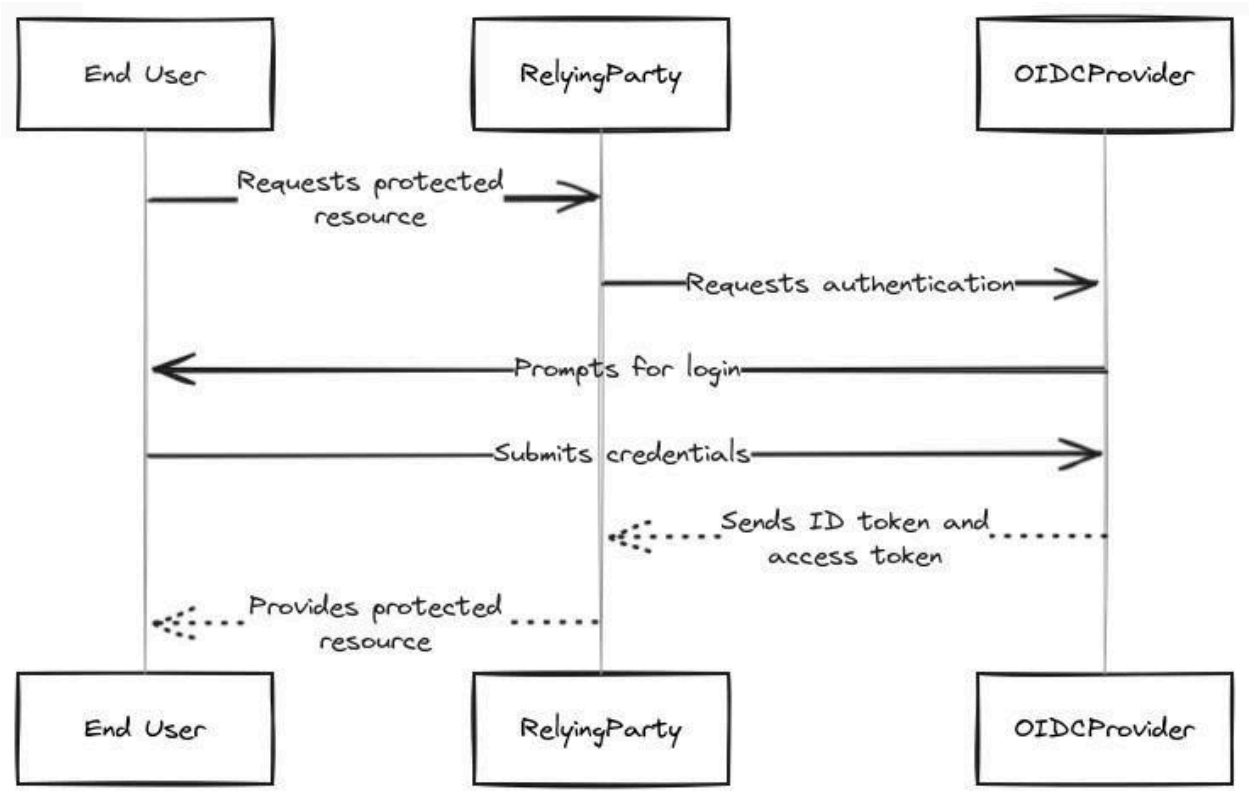
🏗️ OIDC Flow



Automate GitHub deployment with GitHub aws Actions







🚀 Step-by-Step Setup

1️⃣ Create OIDC Provider in AWS

Run once:

```
aws iam create-open-id-connect-provider \
  --url https://token.actions.githubusercontent.com \
  --client-id-list sts.amazonaws.com \
  --thumbprint-list 6938fd4d98bab03faadb97b34396831e3780aea1
```

2️⃣ Create IAM Role for GitHub

✅ Trust Policy (VERY IMPORTANT)

Replace:

- `<ACCOUNT_ID>`
- `<GITHUB_USERNAME>`
- `<REPO_NAME>`

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": {
        "Federated":
"arn:aws:iam::<ACCOUNT_ID>:oidc-provider/token.actions.githubusercontent.com"
      },
      "Action": "sts:AssumeRoleWithWebIdentity",
      "Condition": {
        "StringEquals": {
          "token.actions.githubusercontent.com:aud": "sts.amazonaws.com"
        },
        "StringLike": {
          "token.actions.githubusercontent.com:sub":
"repo:<GITHUB_USERNAME>/<REPO_NAME>:ref:refs/heads/main"
        }
      }
    }
  ]
}
```

3️⃣ Attach Permissions to Role

For your project:

```
{
  "Effect": "Allow",
  "Action": [
    "lambda:*",
    "apigateway:*",
    "dynamodb:*",
    "codedeploy:*",
    "cloudwatch:*",
    "logs:*",
    "iam:PassRole",
    "secretsmanager:*",
    "ec2:*"
  ],
  "Resource": "*"
}
```

👉 Later you can restrict this (least privilege)

4️⃣ Add Role ARN to GitHub

No secrets needed 👍

Just use it directly in pipeline:

role-to-assume: arn:aws:iam::<ACCOUNT_ID>:role/github-actions-role

---

5️⃣ Update GitHub Actions (Already Done)

✅ This is the key block:

```
- name: Configure AWS
  uses: aws-actions/configure-aws-credentials@v2
  with:
    role-to-assume: arn:aws:iam::<ACCOUNT_ID>:role/github-actions-role
    aws-region: us-east-1
```

6️⃣ Add Permissions in Workflow

At top of YAML:

```
permissions:
  id-token: write
  contents: read
```

👉 REQUIRED for OIDC

🧪 How to Verify It Works

Push code:

```
git push
```

Then in logs you should see:

```
Assuming role with OIDC
```

❗ Common Mistakes

❌ Wrong repo name

repo:user/repo

Must match exactly.

❌ Missing permissions

```
permissions:
  id-token: write
```

❌ Wrong branch

If using `main`, trust policy must match:

"ref:refs/heads/main"

❌ Using AWS keys + OIDC together

👉 Remove keys completely

🔐 Security Best Practices

Restrict to branch

"repo:your-org/your-repo:ref:refs/heads/main"

Restrict to environment (advanced)

"repo:your-org/your-repo:environment:prod"